Cambridge (CIE) IGCSE Computer Science

Encryption

Contents

Your notes

Encryption

Your notes

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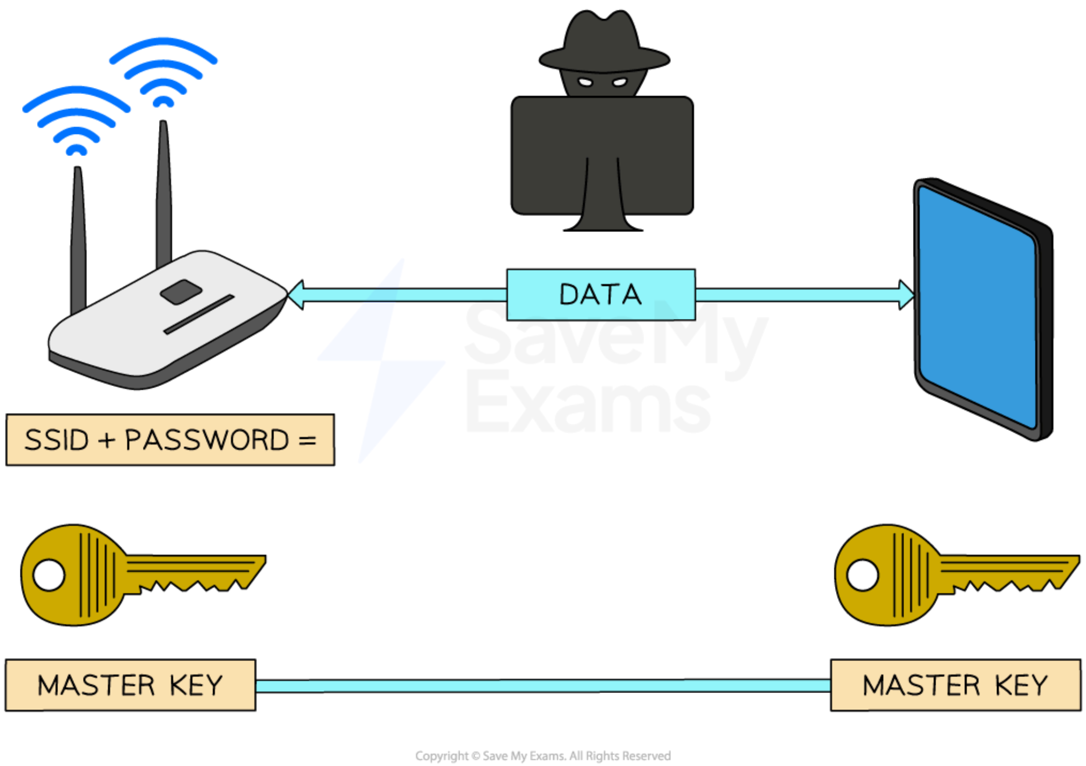
What is encryption?

- Encryption is a method of **scrambling data** before being transmitted across a network
- Encryption helps to protect the contents from unauthorised access by **making data meaningless**
- While encryption is important on both **wired** and **wireless** networks, it's even more critical on wireless networks due to the data being transmitted over radio waves, making it easy to **intercept**

How is wireless data encrypted?

- Wireless networks are identified by a 'Service Set Identifier' (**SSID**) which along with a password is used to create a '**master key**'
- When devices connect to the **same wireless network** using the SSID and password they are given a copy of the master key
- The master key is used to **encrypt** data into '**cipher text**', before being transmitted
- The receiver uses the same master key to **decrypt** the cipher text back to '**plain text**'
- To guarantee the security of data, the **master key is never transmitted**. Without it, any intercepted data is rendered useless
- Wireless networks use dedicated **protocols** like **WPA2** specifically designed for **Wi-Fi** security

Your notes



How is wired data encrypted?

- Wired networks are encrypted in a very similar way to a wireless network, using a **master key to encrypt** data and the **same key to decrypt** data
- Encryption on a wired network differs slightly as it is often left to **individual applications to decide** how encryption is used, for example **HTTPS**

Symmetric & Asymmetric Encryption

- Encryption relies on the **use of a key**
- A key is a **binary string** of a certain length that when applied to an encryption algorithm can **encrypt** plaintext information and **decrypt** ciphertext
  - **Plaintext** is the name for data before it is encrypted
  - **Ciphertext** is the name for data after it is encrypted
- Keys can vary in size and **act like passwords**, enabling people to protect information

What is symmetric encryption?

- Symmetric encryption is when both the sender and receiver are given **an identical secret key** which can be used to **encrypt** or **decrypt** information
- If a hacker **gains access to the key,** then they can **decrypt** intercepted information
- The secret key can be shared with the receiver without sending it electronically:
  - Both parties could **verbally share** the key in person
  - Both parties may use **standard postage mail** to share the key
  - An **algorithm may be used to calculate the key** by sharing secret non-key information

What is asymmetric encryption?

- Asymmetric encryption is when two keys are used, a public and private key
  - **Public key:** a key known to everyone
  - **Private key:** a key known only to the receiver
- **Both keys are needed** to encrypt and decrypt information
- Asymmetric encryption works as follows:
  - **Person A** uses a **public key** to **encrypt their message**
  - **Person A sends their message** over the network or internet
  - **Person B decrypts the message** using their **secret private key**
- **Only one private key can be used** to decrypt a message and it is **not sent over the internet** like a symmetric key
- **Keys can be large**, a key using 100 bits would generate 1,267,650,600,228,229,401,496,703,205,376 different combinations
- Large keys are **near impossible for a hacker to guess**

Worked Example

Complete the sentences about symmetric encryption. Use the terms from the list. Some of the terms in the list will not be used. You should only use a term once.

algorithm

Your notes

Your notes

cipher
copied
delete
key
plain
private
public
standard
stolen
understood
unreadable

The data before encryption is known as ______ text. To scramble the data, an encryption ______, which is a type of ______, is used. The data after encryption is known as ______ text. Encryption prevents the data from being ______ by a hacker.

[5]

Answer

One mark for each correct term in the correct place:

The data before encryption is known as plain text. To scramble the data, an encryption algorithm/key, which is a type of key/algorithm, is used. The data after encryption is known as cipher text. Encryption prevents the data from being understood by a hacker.